

H-NeiFi: Non-Invasive and Consensus-Efficient Multi-Agent Opinion Guidance

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Abstract

Social media's openness fosters opinion exchange but complicates guiding users toward global consensus. Existing approaches are often invasive, modifying opinions or forcing cross-group interactions, undermining autonomy and triggering resistance. They also lack long-term planning, risking macro-level polarization despite local agreement. To address this, we propose H-NeiFi, a hierarchical, non-invasive opinion guidance framework. It models experts and non-experts separately and introduces a neighbor filtering mechanism that adaptively shapes communication without altering user opinions. Using multi-agent reinforcement learning with a long-term reward, H-NeiFi optimizes information flow while preserving interaction autonomy. Experiments show that it accelerates consensus by 22.0% to 30.7% and achieves global convergence even without experts, providing a natural and efficient paradigm for social network governance. Our code is available at: <https://github.com/shijunguo44/H-NeiFi>.

CCS Concepts

• **Computing methodologies** → **Multi-agent planning**; • **Applied computing** → **Sociology**; • **Information systems** → **Process control systems**.

Keywords

Opinion Dynamics, Multi-agent System, Reinforcement Learning

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1 Introduction

With the rapid growth of social media, an increasing number of users exhibit diverse preferences across various topics [27] and actively engage in online social interactions to exchange and update opinions [23]. During this interaction process, phenomena such as consensus formation, disagreement, and polarization often emerge. A significant body of research on classical opinion dynamics models [15, 17] has been extended to address domain-specific applications [21, 43], covering topics like opinion diffusion [33], opinion prediction [32], and opinion maximization [26].

Social media platforms often recommend content based on user preferences to enhance engagement [18]. Yet this practice can narrow users' exposure to diverse viewpoints, reinforcing prior beliefs and fostering the echo chamber effect [13], which in turn aggravates social biases and fragmentation [38]. A typical manifestation is the emergence of isolated clusters, as illustrated in Figure 1b. Motivated by these concerns, recent work on opinion consensus [1, 36] has gained momentum. Existing studies pursue different intervention strategies, including encouraging communication between users with large disagreements [46, 47], amplifying the influence of polarized opinions [5], and designing goal-oriented control functions [34].

Despite these advances, deploying opinion-consensus mechanisms in real-world platforms raises several practical challenges that require careful consideration. In particular, existing methods still exhibit notable limitations, mainly due to two reasons:

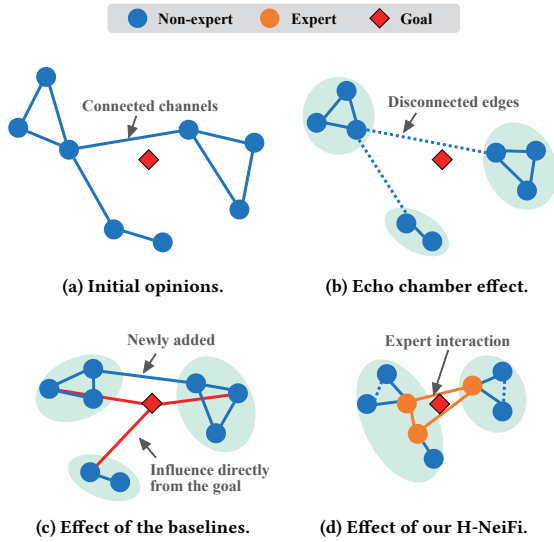


Figure 1: Illustration of opinion evolution in a social network. (a) illustrates the initial opinion distribution, (b) demonstrates isolated clusters formed by the echo chamber effect. During the evolution process, the convergence of existing opinion baselines and our method is shown in (c) and (d). The baselines rely on direct influences, resulting in localized consensus. In contrast, our H-NeiFi achieves global consensus through hierarchical communication patterns and a non-invasive neighbor filtering method.

Invasive: (i) Some approaches [46, 47] add connections between users with different opinions and enforce interactions between them, while others [9, 37] introduce external opinions based on user opinion distributions to mitigate the echo chamber effect. An example of these newly imposed connections is shown in Figure 1c. This intervention disrupts the social autonomy of user selection. As a result, exposure to belief-conflicting content (e.g., political values) can induce psychological reactance, paradoxically reinforcing users’ original positions [35]. (ii) Some approaches [8, 41] directly introduce influence from the opinion goal to facilitate consensus. In these approaches, users with significant opinion differences are directly controlled by a goal-oriented function, as shown in Figure 1c. However, opinion formation is inherently a continuous evolution process based on individual cognition and social interaction. The discrete numerical modifications imposed by these methods lack a solid behavioral science foundation and may trigger psychological resistance mechanisms, such as the backfire effect [10].

Ineffective: Existing guidance methods [20, 25] often overlook that public opinion interaction constitutes a long-term, complex dynamic process. As a result, they drive users to form stronger connections with opinion-aligned peers [29], over-optimizing local cohesion risks creating self-reinforcing echo chambers that amplify polarization and foster extreme communities at the macroscopic level [6]. Such localized consensus paradoxically undermines global harmony and establishes a detrimental feedback loop. Effective strategies must, therefore, balance immediate efficiency with

long-term stability, avoiding self-reinforcing echo chambers while steering consensus toward socially aligned outcomes.

Based on the above analysis, we propose H-NeiFi, a novel opinion guidance framework that integrates hierarchical user behavior patterns with a non-invasive neighbor filtering approach. H-NeiFi aims to guide users toward an inclusive consensus aligned with a predefined opinion goal [19]. We categorize users as experts or non-experts: experts interact only among themselves, while non-experts engage with peers and assimilate expert opinions. Opinion dynamics are modeled using a bounded confidence model [2] extended with stubbornness [42] and attention goals.

For guidance, H-NeiFi performs pruning-based neighbor filtering via two paradigms: Promotion Communication Pattern (PCP) and Attention Communication Pattern (ACP). PCP speeds up propagation by linking low-divergence experts consistent with the global goal. ACP adopts a Multi-Agent Reinforcement Learning (MARL) [24] formulation, treating non-experts as agents to trade off communication efficiency and long-term consensus. Importantly, H-NeiFi neither alters user opinions nor injects external nodes; it optimizes information diffusion by dynamically adjusting the visibility of existing neighbors (“neighbor filtering”). This yields two benefits: (i) By selectively hiding non-goal-oriented neighbors (e.g., temporarily masking irrelevant connections), it feels more natural and is less likely to provoke resistance from users, as they are generally less sensitive to the disappearance of relationships during routine interactions [16]. (ii) interventions remain local, avoiding trust costs from unfamiliar contacts.

Our approach H-NeiFi accelerates consensus formation by 22.0–30.7% on average during opinion guidance. Notably, it maintains robust performance even without expert intervention, consistently achieving unified consensus. As visually demonstrated in Figure 1d, H-NeiFi outperforms existing methods across all key metrics.

In summary, our main contributions are three-fold:

- We propose H-NeiFi, a non-invasive opinion-guidance framework that leverages hierarchical interactions (experts vs. non-experts) to drive goal-oriented global consensus.
- We introduce a MARL-based dynamic neighbor filtering scheme that adaptively shapes local communication over long horizons, reducing fragmentation without invasive control.
- Experiments across diverse scenarios and scales demonstrate the effectiveness and superiority of our approach.

2 Related Work

In opinion guidance studies, where users are directed to converge toward a specific goal, external interventions are commonly used to influence opinion changes. Kurz [22] introduced “strategic agents”, whose opinions directly influence others, while Dietrich et al. [14] proposed a leadership model with bounded confidence, where users align with a leader to reach the target opinion. Cicolani et al. [12] explored consensus in scenarios where agents are influenced by a common influencer, even with non-universal interactions and time delays. In these approaches, the leader’s opinion is controlled and directly shapes followers’ opinions.

Some methods use RL to enhance consensus. Wang et al. [40] introduced an RL-based framework for adaptive opinion management, guiding users toward a target opinion with minimal cost. Borkar

and Reiffers-Masson [4] applied RL to identify influential users and navigate complex network topologies. Unlike these cost-driven methods, our work prioritizes consensus efficiency optimization.

Recent studies by Nugent et al. [31] and Bolzern et al. [3] have examined the role of interaction intensity in opinion guidance. While these works show that altering interaction behaviors can influence opinions, they overlook hierarchical relationships among users and the efficiency of the opinion guidance process.

3 Preliminaries

System Model and Problem Statement. We consider a social network with n experts $E = \{e_1, \dots, e_n\}$ and m non-experts $V = \{v_1, \dots, v_m\}$, whose opinions at time k are denoted by $x_{e_i}(k)$ and $x_{v_i}(k)$, respectively. Without loss of generality, we assume ordered initial opinions: $x_{v_1}(0) \leq \dots \leq x_{v_m}(0)$ (similarly for experts). Experts maintain fixed influence due to institutional authority. The system seeks non-invasive consensus aligned with the global goal U within finite time T , terminating at the earliest $k \leq T$ satisfying:

$$\max_{v_i, v_j \in V} |x_{v_i}(k) - x_{v_j}(k)| \leq \omega \quad \text{and} \quad x_{v_i}(k) \rightarrow U \quad \forall v_i \in V, \quad (1)$$

where ω is the consensus threshold and all opinions lie in $[x_{\min}, x_{\max}]$. H-NeiFi's neighbor filtering provides guidance to align the consensus value with the global goal U as closely as possible under this condition.

MARL under Opinion Dynamics Scenarios. Markov Decision Processes (MDPs) are the basis of modeling and analysis in MARL. According to the analysis in the previous section, we extend the definition of MDPs in MARL proposed by Zhang et al. [44], and defined a new MDP for the opinion dynamics under social networks in Eq. (2). The brief introduction is given as follows:

$$\left(\{S_{v_i}\}, \{A_{v_i}\}, P, \{R_{v_i}\}, \{L(k)\}_{k \geq 0} \in \mathbb{R}^{m \times m} \right), \quad (2)$$

where $\{S_{v_i}\}$ represents the local state space perceived by agent v_i , $\{A_{v_i}\}$ represents the local action space made by agent v_i , $\{L(k)\}_{k \geq 0}$ indicates the communication topology matrix of the system at time k . If there exists a communication channel from agent v_i to agent v_j , then l_{ij} is 1, otherwise, l_{ij} is 0. Then, the local reward function obtained by each agent i is $R_{v_i} : S \times A \rightarrow R$, the state transition probability of MDP is $P : S \times A \times S \rightarrow [0, 1]$.

The Markov decision process defined in this paper can be expressed as follows: agent v_i perceives the local observation state $S_{v_i}(k)$ at time k . Then, v_i get the action probability distribution $A_{v_i}(k)$ according to the shared policy parameters π_θ . Based on the $A_{v_i}(k)$, we update the communication matrix $L(k)$. Meanwhile, v_i receives its local instant rewards $r_{v_i}(k)$ based on the reward function. We denote the trajectory of agent v_i during a training round as $\tau_{v_i} = (S_{v_i}(0), A_{v_i}(0), r_{v_i}(0), \dots, S_{v_i}(H), A_{v_i}(H), r_{v_i}(H))$, where H is the length of the trajectory. In the process of interaction with the environment, agents constantly learn and adjust their strategy function $\pi_\theta(A_{v_i}(k)|S_{v_i}(k))$ to maximize the expected cumulative $\bar{R}_\theta$ return in the future:

$$\bar{R}_\theta = \sum_{\tau_{v_i}} P(\tau_{v_i}|\theta) R(\tau_{v_i}), \quad (3)$$

where $R(\tau_{v_i})$ is the total reward obtained from τ_{v_i} , and $P(\tau_{v_i}|\theta)$ is the probability that the trajectory occurs.

4 Methodology

4.1 Overall Framework

To address the issue of guiding users' opinions towards achieving an inclusive consensus on the goal through neighbor filtering, we propose H-NeiFi for multi-agent opinion guidance (Figure 2). It consists of three phases: environment initialization, user communication, and policy exploitation and update.

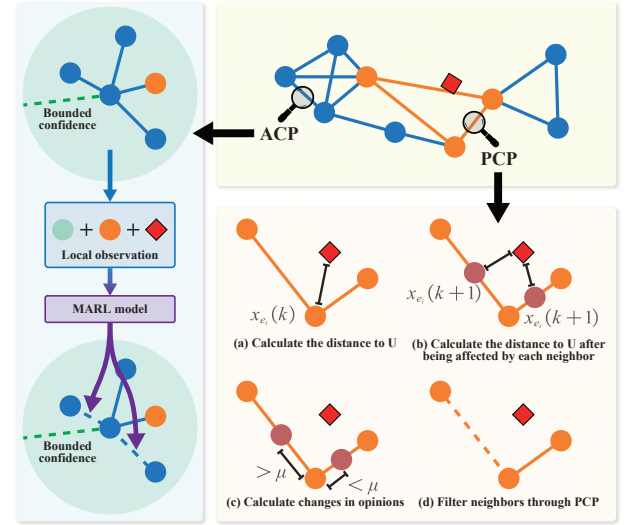


Figure 2: H-NeiFi framework. At the start of each training round, a global goal is set and users are categorized into non-experts and experts. PCP filters neighbors for experts based on the minimum disagreement, while ACP filters neighbors for non-experts based on MARL for long-term planning.

Environment Initialization Phase. In each training round, we set a global consensus goal and categorized the users into two roles: experts and non-experts. Each role follows a distinct opinion dynamics model that accounts for different levels of stubbornness. **User Communication Phase.** Users engage in role-specific interaction patterns: PCP assigns experts' neighbors based on minimal opinion divergence, propagating global goal influence to non-experts through expert consensus. Meanwhile, ACP dynamically prunes communication channels via MARL-driven balancing of local stability and global consensus, optimizing non-expert interactions through long-term planning strategies. The communication process continues until it either reaches the maximum time step T or achieves the consensus threshold ω .

Policy Exploitation and Updating Phase. First, the reward obtained by each non-expert in the round is calculated based on the reward function. Then, the adaptive attention network within the ACP is trained and optimized using a policy gradient algorithm, learning a policy function that maximizes the expected reward.

4.2 PCP for Experts

Expert users on social media exhibit stable opinions and are less susceptible to shifts than non-experts [7, 45]. To model this, we

employ a stubbornness-based opinion dynamics model as the foundational update rule for experts. The opinion of expert e_i at the next time step is given by:

$$x_{e_i}(k+1) = \phi_{e_i} \cdot x_{e_i}(k) + (1 - \phi_{e_i}) \cdot \frac{\sum_{j \in N_{e_i}(k)} x_{e_j}(k)}{|N_{e_i}(k)|}, \quad (4)$$

where $\phi_{e_i} \in [0, 1]$ represents the stubbornness level of expert e_i , and $N_{e_i}(k)$ is the set of neighbors that e_i interacts with at time k . *Base Model Specification:* Without external guidance, we model the expert community as a tightly-knit group where everyone is aware of each other's stance. This reflects typical behavior among high-profile individuals (e.g., academics, influencers) within a domain. Therefore, the default neighbor set in the base model is defined by global mutual awareness: $N_{e_i}(k) = E \setminus e_i$ for all experts. This models a fully connected communication topology among experts.

The core of our non-invasive guidance lies in controlling the composition of $N_{e_i}(k)$. We propose the Promotion Communication Pattern (PCP) algorithm to adaptively filter the neighbor set for each expert, steering their interactions toward the global goal U without altering their core update rule (Eq. 4).

The PCP algorithm operates as follows. First, it identifies the candidate neighbor set $G_{e_i}(k)$ from all experts, comprising those who can facilitate e_i 's convergence:

$$G_{e_i}(k) = \left\{ e_j \in E \mid \left| \frac{x_{e_i}(k) + x_{e_j}(k)}{2} - U \right| \leq |x_{e_i}(k) - U| \right\}. \quad (5)$$

Subsequently, PCP constructs the final interaction set $\hat{G}_{e_i}(k) \subseteq G_{e_i}(k)$ through an iterative process:

- Step 1:** Select the nearest neighbour e_j from $G_{e_i}(k)$ and add it to $\hat{G}_{e_i}(k)$.
- Step 2:** Calculate the prospective new opinion $x_{e_i}(k+1)$ using Eq. 4 with $N_{e_i}(k) = \hat{G}_{e_i}(k)$.
- Step 3:** If $|x_{e_i}(k+1) - x_{e_i}(k)| \geq \mu$, the selection is complete. Otherwise, return to Step 1.

Parameter μ sets the threshold for expert opinion shifts because, despite their higher stubbornness, experts exhibit greater disagreement within the same opinion range. Consequently, their opinions can drift more significantly during exchange. Here, the parameter μ serves as a stability threshold, preventing excessive opinion fluctuations and ensuring that experts provide consistent and reliable guidance to non-experts.

The updated opinions of experts subsequently reshape the social structure among non-experts. Specifically, non-experts are dynamically partitioned into subgroups based on their opinion proximity to experts. Each non-expert agent v_i aligns with the nearest expert, denoted $LE_{v_i}(k)$, forming a follower relationship. Thus, each subgroup comprises one expert and its associated non-experts.

This subgroup architecture is inherently dynamic. When the average opinion difference between the leading experts of two subgroups falls below a merging threshold, the two subgroups consolidate into a new, larger group. This mechanism enables the system to first foster local consensus within clusters before progressively merging toward global consensus, guided cohesively by the ongoing convergence of expert opinions under PCP.

4.3 ACP for Non-experts

Unlike opinion-stable experts, non-experts exhibit greater flexibility and are more readily influenced in social networks. To model their opinion evolution, we first define the foundational stubbornness-based dynamics model for non-experts:

$$x_{v_i}(k+1) = \begin{cases} \phi_{v_i} x_{v_i}(k) + (1 - \phi_{v_i}) C_{v_i}(k), & N_{v_i}(k) \neq \emptyset, \\ x_{v_i}(k), & N_{v_i}(k) = \emptyset, \end{cases} \quad (6)$$

where ϕ_{v_i} represents the stubbornness level of non-expert v_i , and $C_{v_i}(k)$ represents the weighted opinion influence received from both neighbors and followed experts at time k . *Base Model Specification:* In the absence of guidance, each non-expert v_i interacts with all agents within its bounded confidence range r_c . Thus, the default neighbor set for the base model is defined as $N_{v_i}(k) = \{ v_j \in V \mid |x_{v_i}(k) - x_{v_j}(k)| < r_c \}$.

To improve consensus convergence, we introduce the Attention Communication Pattern (ACP) as an external guidance mechanism. Instead of modifying the update rule, ACP dynamically filters $N_{v_i}(k)$ to promote global alignment while preserving local cohesion. Since each filtering step changes the social network and thereby influences subsequent opinions and future filtering, neighbor selection becomes a sequential decision-making problem.

This problem can be formulated as a MARL task. Each non-expert acts as an agent that learns a neighbor-selection policy to maximize long-term cumulative reward, reflecting local and global consensus goals. The policy network must address two challenges: (1) *variable-sized inputs* caused by time-varying neighbor counts, and (2) *temporal dependencies* arising from past interactions. We therefore adopt a bidirectional LSTM architecture that encodes dynamic-length neighbor lists and maintains state information, enabling context-aware neighbor selection.

The local goal $O_{v_i}(k)$ conveys expert guidance to non-experts and is defined as the average opinion of v_i 's expert neighbors:

$$O_{v_i}(k) = \frac{1}{|LE_{v_i}(k)|} \sum_{e_j \in LE_{v_i}(k)} x_{e_j}(k). \quad (7)$$

Given $O_{v_i}(k)$ and the global goal U , we construct the input features by measuring absolute deviations to both goals. For agent v_i , we define:

$$d_{v_i}^O(k) = |x_{v_i}(k) - O_{v_i}(k)|, \quad d_{v_i}^U(k) = |x_{v_i}(k) - U|, \quad (8)$$

and any value below 0.1 is clamped to 0.1 for numerical stability. For each neighbor $v_j \in N_{v_i}(k)$, we further define the directed distances:

$$d_{v_j \rightarrow i}^O(k) = |x_{v_j}(k) - O_{v_i}(k)|, \quad d_{v_j \rightarrow i}^U(k) = |x_{v_j}(k) - U|. \quad (9)$$

We then represent v_i 's neighborhood at time k as a set of 4-dimensional feature vectors and randomly shuffle them to reduce ordering bias:

$$S_{v_i}(k) = \text{shuffle} \left(\left\{ \left[\frac{d_{v_j \rightarrow i}^O(k)}{d_{v_i}^O(k)}, \frac{d_{v_j \rightarrow i}^U(k)}{d_{v_i}^U(k)}, d_{v_i}^O(k), d_{v_i}^U(k) \right]^\top \right\}_{v_j} \right), \quad (10)$$

where the value less than 1 in the first (second) dimension indicates that neighbor v_j is closer than v_i to the local (global) goal.

Conditioned on this representation, the bidirectional LSTM outputs neighbor-selection preferences, $A_{v_i}(k) = \text{LSTM}(S_{v_i}(k))$. Applying a Softmax yields the normalized preference scores $\{\pi_{\theta_i}^j(k)\}$ for $v_j \in N_{v_i}(k)$, where $\pi_{\theta_i}^j(k)$ denotes agent v_i 's relative preference for neighbor v_j at time k (summing to 1 over the current neighbor set). We convert these preferences into binary link decisions via a relative-thresholding rule: we set $\hat{\pi}_{\theta_i}(k)$ to the mean of $\{\pi_{\theta_i}^j(k)\}$ and use it as the threshold, and update the communication topology matrix $L(k) = [l_{ij}(k)]$ as:

$$l_{ij}(k) = \begin{cases} 1, & v_j \in N_{v_i}(k) \text{ and } \pi_{\theta_i}^j(k) \geq \hat{\pi}_{\theta_i}(k), \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

Consequently, v_i retains neighbors with above-average preference, enabling adaptive filtering under varying neighbor sizes. During training, we flip the keep/drop decisions with probability ε to promote exploration.

After ACP filtering, agents interact with their dynamically selected neighbors. The assimilated opinion $C_{v_i}(k)$ is computed as:

$$C_{v_i}(k) = p \cdot O_{v_i}(k) + q \cdot \frac{\sum_{v_j \in V} l_{ij}(k) \cdot x_{v_j}(k)}{\sum_{v_j \in V} l_{ij}(k)}, \quad (12)$$

which combines expert guidance (p -weighted) and peer influence (q -weighted) within the adapted communication topology.

The non-expert opinion is then updated according to the base dynamics model (Eq. 6) using this filtered input, completing one iteration of the guided opinion evolution process.

4.4 Reward Design and MARL Optimization

To enable non-expert users to balance local and global consensus goals during long-term opinion evolution, we design a dual-reward mechanism that integrates two complementary goals, defined as:

$$r_{v_i}(k) = g_1(\cdot) + g_2(\cdot), \quad (13)$$

where this mechanism guides non-experts toward global goal convergence while maintaining local group connectivity, thereby preventing premature fragmentation caused by limited confidence. Given that the immediate impacts of communication channels may manifest only after multiple steps, we employ policy gradient optimization to associate current actions with long-term benefits by maximizing cumulative rewards, effectively mitigating negative effects from myopic decision-making.

Local Consensus Reward. $g_1(\cdot)$ guides agents toward local consensus goal by ensuring users maintain communication ranges with their subgroups, thereby preventing edge users from isolation:

$$g_1 = \begin{cases} \frac{d_{v_i}^O(k) - d_{v_i}^O(k+1)}{d_{v_i}^O(k)}, & d_{v_i}^O(k) \geq \xi_l, \\ \frac{\xi_l - d_{v_i}^O(k+1)}{\xi_l}, & d_{v_i}^O(k) < \xi_l, \end{cases} \quad (14)$$

where ξ_l denotes the stabilization threshold from the agent to the local goal.

The local consensus reward activates when $d_{v_i}^O(k) \geq \xi_l$: positive g_1 yields distance-proportional rewards, while negative g_1 incurs distance-proportional penalties. Meanwhile, the stabilization reward triggers when $d_{v_i}^O(k) < \xi_l$, rewarding maintained proximity ($d_{v_i}^O(k+1) \leq \xi_l$) and penalizing threshold breach ($d_{v_i}^O(k+1) > \xi_l$).

Global Consensus Reward. $g_2(\cdot)$ drives agents to converge to the global consensus goal, prioritizing long-term consensus guarantees over short-term cohesion and thereby preventing subgroup fragmentation:

$$g_2 = \begin{cases} \frac{d_{v_i}^U(k) - d_{v_i}^U(k+1)}{d_{v_i}^U(k)}, & d_{v_i}^U(k) \geq \xi_g, \\ \frac{\xi_g - d_{v_i}^U(k+1)}{\xi_g}, & d_{v_i}^U(k) < \xi_g, \end{cases} \quad (15)$$

where ξ_g represents the stabilization threshold for the distance to the global goal. While the meaning of g_2 is similar to g_1 , its objective shifts from local to global.

Our objective is to maximize the expectation of $\bar{R}_\theta$. We sample the trajectories of all agents in a round and estimate the gradient by averaging the approximate values:

$$\nabla_\theta \bar{R}_\theta \approx \frac{1}{m} \sum_{i=1}^m \sum_{k=0}^H \nabla_\theta \log \pi_\theta(A_{v_i}(k) | S_{v_i}(k)) \cdot R(\tau_{v_i}). \quad (16)$$

Then, we use the gradient descent method with a learning rate α to update the policy parameters θ :

$$\theta \leftarrow \theta + \alpha \nabla_\theta \bar{R}_\theta. \quad (17)$$

5 Experiment

5.1 Experimental Settings

In this section, extensive experiments were conducted to assess the effectiveness of H-NeiFi in guiding consensus toward a target opinion. However, acquiring real-world dynamic process data for closed-loop public opinion guidance remains challenging, as most publicly available graph datasets contain static opinion data, which fails to capture user evolution after intervention. To address this, many studies [8, 12, 30] rely on synthetic datasets, which allow for better control of experimental variables and deeper analysis of model behavior. This study similarly uses synthetic datasets to ensure experimental reproducibility and the generalizability of findings. Experiments were conducted across various benchmarks, covering different opinion ranges and agent counts.

To validate H-NeiFi's efficacy and superiority, we evaluate the performance using the following metrics:

- **Consensus Cluster (CC):** Number of opinion clusters formed at consensus threshold or maximum steps. Lower CC implies greater consensus.
- **Consensus Step (CS):** Steps required for the system to reach the consensus threshold. Lower CS indicates higher guidance efficiency.
- **System Consensus Deviation (SCD):** Average deviation of agent opinions from the global goal at termination. Lower SCD indicates higher guidance accuracy.

5.2 Comparison methods

We compare the opinion guidance performance of H-NeiFi with the following baseline methods:

- **Common-Neighbor (CNR)** [39]: Randomly selects users outside the current user's perception range and adds them to their neighbors.

Table 1: Statistics of consistency comparison experiment. The best performances are highlighted with green, and our results are bolded.

Init. range	m	Method	CC ↓	CS ↓
[0, 4]	20	PWA	1	13.2 ± 0.36
		CNR	1	16.3 ± 0.61
		GP	1	14.3 ± 0.61
		H-NeiFi	1	13.0 ± 0.40
[0, 4]	40	PWA	1	13.8 ± 0.56
		CNR	1	17.7 ± 0.41
		GP	1	14.5 ± 0.65
		H-NeiFi	1	13.1 ± 0.29
[0, 8]	40	PWA	3	15.1 ± 0.29
		CNR	1.5 ± 0.25	34.5 ± 0.25
		GP	3	15.6 ± 0.24
		H-NeiFi	1	20.5 ± 1.25
[0, 8]	80	PWA	3	16.2 ± 0.36
		CNR	2.7 ± 0.21	35 ± 0
		GP	3	17.3 ± 0.61
		H-NeiFi	1	22.0 ± 2.80

- Group-Pressure (GP) [11]: Sets group pressure, causing the opinions of certain users to be directly influenced by external group pressures.
- Position Weight Allocation (PWA) [28]: Assigns weights to each user's neighbors based on their proximity to the average opinion in the current environment. Neighbors whose opinions are closer to the average receive higher weights.

While CNR and GP are direct intervention methods, PWA does not directly alter user communication but instead allocates weights based solely on the current environment without considering long-term planning. To ensure a fair comparison, all experiments employ a unified opinion dynamics model based on our proposed hierarchical framework.

5.3 Comparative Experiments of Consensus Guidance without Experts

To verify the effectiveness of our method in facilitating consensus, we first test with uniformly distributed agent opinions. Since traditional methods typically converge to the opinion midpoint, U is set to be the midpoint of the opinion range. An expert with fixed opinion provides only ACP's local goal ($p = 0, q = 1$). All results present the mean and variance across multiple randomized trials.

Table 1 summarizes comparative consensus results under identical conditions. Our method consistently achieves global consensus within specified steps across all tested environments, opinion ranges, and agent counts. Increasing agents within the same opinion range has little effect on consensus steps or clusters. However, expanding the opinion range noticeably increases consensus steps and reduces quality, suggesting higher consensus costs with greater opinion divergence.

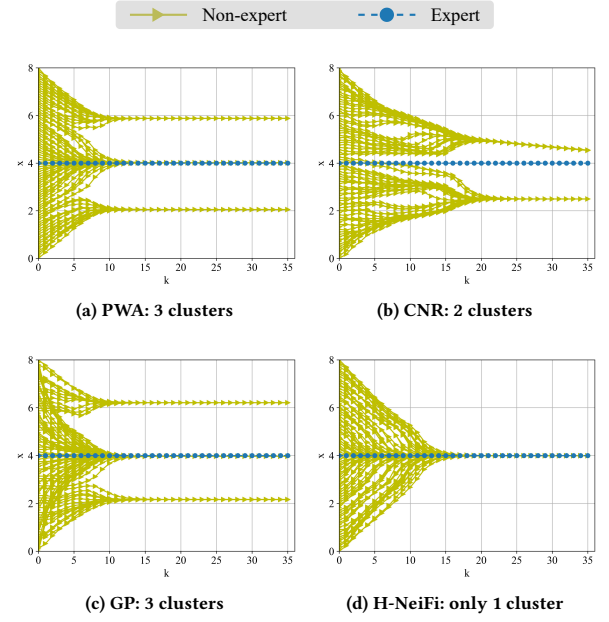


Figure 3: Illustration of opinion evolution processes without expert influence ($x_{v_i}(0) \in [0, 8]$ and $m = 80$).

Within the opinion range of $[0, 8]$, we observe that PWA and GP require significantly fewer steps to reach consistency but result in a higher number of consensus clusters. In contrast, CNR, though needing more steps and sometimes failing to achieve full consensus within the maximum allowed steps, shows a clear trend toward consensus and is likely to reach global consensus in the future. As shown in Figure 3, PWA and GP fall into local consensus prematurely. Similarly, while CNR shows a consensus trend, it also first reaches local consensus, requiring additional time to achieve global consistency. In comparison, H-NeiFi, though not the fastest in terms of consensus speed, avoids local consensus and ultimately achieves global consensus by trading off some speed for accuracy.

5.4 Comparative Experiments of Opinion Guidance with Experts

We compared the effectiveness of opinion guidance across various opinion ranges and agent numbers. While most existing methods rely on a single expert to directly influence non-experts, our framework incorporates multiple experts. To ensure a fair comparison, we applied our proposed hierarchical framework to PWA, GP, and CNR. In experiments assessing expert opinion impact, we set $p = 0.1$ and $q = 0.9$.

By comparing Tables 1 and 2, it is evident that with the inclusion of expert influence, all methods converge into a single cluster within the maximum allowed steps. Our method requires the fewest consensus steps across various environments and minimizes the SCD metric when the system reaches consensus, indicating that it is the closest to achieving the global consensus goal.

From Table 2, we observe that the consensus steps are longer in the range $[0, 4]$ compared to the no-expert scenario. This is because the global goal is not at the opinion range midpoint. Under experts'

Table 2: Statistics of opinion control with experts. In the Init. range $[0, 4]$, $X_{E_i}(0) = \{1, 3\}$, $U = 1.5$. In the Init. range $[0, 8]$, $X_{E_i}(0) = \{1, 3, 5, 7\}$, $U = 3$.

Init. range	m	Method	CC ↓	CS ↓	SCD ↓
[0, 4]	20	PWA	1	18.5 ± 4.05	0.258 ± 0.0003
		CNR	1	17.0 ± 0.60	0.171 ± 0.0005
		GP	1	17.3 ± 5.41	0.251 ± 0.0011
		H-NeiFi	1	13.7 ± 0.61	0.050 ± 0.0000
[0, 4]	40	PWA	1	19.4 ± 0.84	0.239 ± 0.0002
		CNR	1	19.5 ± 1.05	0.190 ± 0.0001
		GP	1	21.0 ± 4.60	0.236 ± 0.0001
		H-NeiFi	1	13.8 ± 0.56	0.050 ± 0.0003
[0, 8]	40	PWA	1	34.3 ± 0.81	0.223 ± 0.0002
		CNR	1	27.8 ± 1.36	0.182 ± 0.0024
		GP	1	32.3 ± 5.21	0.150 ± 0.0087
		H-NeiFi	1	24.1 ± 3.49	0.069 ± 0.0009
[0, 8]	80	PWA	1	35 ± 0	0.253 ± 0.0003
		CNR	1	33.5 ± 0.25	0.160 ± 0.0005
		GP	1	35 ± 0	0.377 ± 0.0000
		H-NeiFi	1	25.1 ± 1.69	0.077 ± 0.0001

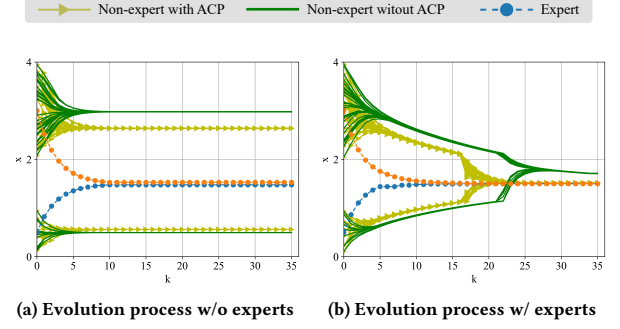
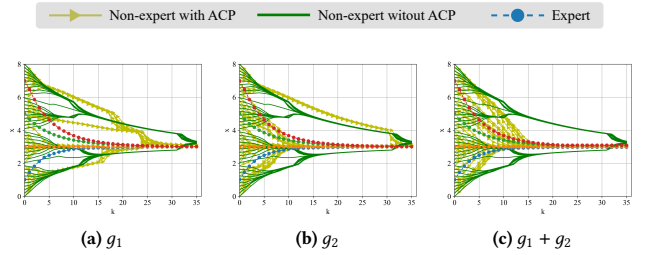
influence, agents adjust further toward the goal, slowing down the overall consensus. While our method is also affected, the change is minimal.

Within the $[0, 8]$ opinion range, all methods eventually reach global consensus, though with more steps. Notably, CNR now converges faster than PWA and GP, a reversal from expert-free scenarios. Without expert guidance, PWA and GP often trap agents in local consensus, requiring prolonged expert intervention. In contrast, CNR’s inherent tendency toward global consensus facilitates easier expert guidance. Our method leverages long-term planning to proactively manage communication, achieving faster global consensus with fewer steps.

5.5 Opinion Evolution Analysis under Unevenly Distributed Environments

To demonstrate method generalizability, we tested under uneven initial opinion distributions: agents initialized randomly in $[0, 1]$ ($m = 10$) and $[2, 4]$ ($m = 30$), with experts at 0.5 and 3. $U = 1.5$ accounts for the agent gap in $[2, 3]$.

In Figure 4a, $p = 0$ and $q = 1$, we observe that under the action of ACP, the consensus result of the agents is relatively closer to the goal. However, due to bounded confidence, they only achieve local consensus and fail to converge fully to the goal. In Figure 4b, $p = 0.1$ and $q = 0.9$, we see that under ACP, non-experts achieve global consensus towards the goal, and the consensus speed is faster compared to scenarios without ACP intervention. Meanwhile, as shown by the green trajectory, when the opinion distance between two groups is less than r_c , the two groups will approach each other and the final result is closer to the one with more agents. However, with ACP intervention, the group with a larger number of agents

**Figure 4: Opinion evolution processes under uneven opinion distributions.****Figure 5: The effectiveness of ACP under different reward combinations.**

does not drag the group with a smaller number of agents away, but both of them converge towards the goal. This demonstrates that ACP can effectively combine the global goal and the local information perceived by the agents to make decisions adaptively.

5.6 Ablation Study of Reward

We conducted experiments under conditions where the stubbornness level of experts and non-experts were fixed. As shown in Figure 5, the global goal is set to 3, there are 40 non-experts and 4 experts, and the opinion range is $[0, 8]$.

When only the local consensus reward g_1 is applied (Figure 5a), non-experts rapidly converge towards the opinions of their followed experts, leading to local consensus and the formation of echo chambers. This results in clusters of non-experts that are too far apart from each other, restricted by bounded confidence limits, preventing inter-cluster communication. Consequently, non-experts can only gradually converge towards the global goal under experts guidance in subsequent steps. However, it is noteworthy that when the distance between clusters falls within the bounded confidence range, consensus towards the global goal is more pronounced compared to scenarios without ACP intervention. At this point, experts have already reached a consensus, making the perceived local goal similar to the global goal for non-experts.

In scenarios where only the global consensus reward g_2 is applied (Figure 5b), non-experts near the global goal quickly reach consensus there under ACP intervention. However, this causes peripheral non-experts to experience an island effect and lose potential communication partners. As a result, although most non-experts reach

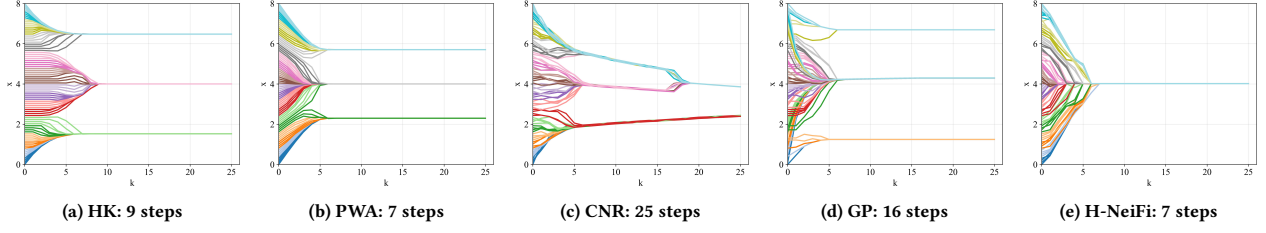


Figure 6: Opinion evolution trajectories under different intervention methods in the HK dynamics model ($x_{v_i}(0) \in [0, 8]$ and $m = 80$).

opinion consensus, those on the periphery converge slowly toward the global goal under expert guidance, significantly reducing global consensus efficiency.

Finally, when combining rewards g_1 and g_2 , as shown in Figure 5c, ACP effectively balances the selection of neighbors based on both local and global goals. For non-experts near the global goal, ACP moderates their consensus speed, preventing peripheral non-experts from falling into the island effect. ACP ensures that a portion of non-experts continuously serve as “bridges” between non-experts, ultimately enhancing the efficiency of achieving global consensus towards the global goal. This demonstrates the effectiveness of our proposed reward combination.

5.7 Evolutionary Analysis under the HK Model

Our proposed hierarchical opinion dynamics model is essentially a variant of the HK model [2]. It extends the original HK model by incorporating user stubbornness and behavioral differences among distinct user roles. Notably, when no experts are present in the environment and all non-experts have a stubbornness value of zero, our model reduces to the classical HK model. To further investigate the generalizability of ACP, we conducted experiments under this equivalent HK configuration. For fair comparison, the baseline opinion dynamics model was also set to the standard HK model.

As shown in Figure 6, when the HK model is employed as the opinion dynamics model, user opinions converge more rapidly. This acceleration occurs because the HK model does not account for user stubbornness; each user fully adopts their neighbors’ opinions during interactions, leading to faster opinion shifts. Under such conditions, users are more prone to form local consensus clusters, often resulting in a fragmented opinion landscape with multiple distinct clusters. In contrast, our proposed ACP guidance mechanism achieves a balance between convergence speed and convergence quality, ensuring rapid yet globally consistent consensus. As demonstrated in Figure 6e, H-NeiFi attains the highest convergence quality, illustrating that even under the classical HK opinion dynamics model, the ACP mechanism effectively facilitates consensus formation among users.

5.8 Efficiency analysis

H-NeiFi operates distributively, requiring each agent access only to local neighbors and the global goal. With E denoting total communication edges, ACP’s per-iteration time complexity is $O(E)$. Crucially, ACP intervention typically terminates before reaching

the theoretical bound ($E = N^2$) as consensus emerges naturally with local communication. Given the sparsity of real-world social networks and ACP’s reliance on local information, it scales feasibly. Per-step GPU processing times for all agents are shown in Table 3.

Table 3: GPU (RTX-3070) Time Scaling by Agent Count

m	Init. range	Time/step (s)
50	[0, 5]	0.0389
100	[0, 5]	0.0851
500	[0, 50]	0.4268
1000	[0, 50]	0.9961
5000	[0, 500]	6.3233

6 Conclusion and Discussion

Social media platforms often amplify polarization despite efforts to boost engagement. Existing opinion guidance approaches usually depend on invasive interventions, such as injecting external opinions or imposing control functions, and they often lack long-term planning while favoring local consensus over global agreement. To overcome these issues, we propose H-NeiFi, a hierarchical neighbor filtering framework that incorporates user roles to steer opinion dynamics in a non-invasive way. H-NeiFi distinguishes experts from non-experts and guides interactions through two adaptive mechanisms: PCP selects expert neighbors based on opinion similarity, and ACP uses MARL to adjust non-experts’ communication ties using both global targets and local context. Experiments demonstrate that H-NeiFi accelerates convergence toward the target consensus and promotes broader opinion harmony. Our study also raises an ethical concern: recommendation algorithms may shape opinions without users’ awareness, for example by selectively concealing content. Although H-NeiFi aims to support constructive consensus, its potential misuse highlights the need to examine how consensus-building mechanisms could be exploited for opinion manipulation, which we leave for future work.

Acknowledgments

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A Derivation of The Policy Gradient

We calculate the gradient of θ in Eq. (3) to maximize the expected value of the objective function $\bar{R}_\theta$:

$$\begin{aligned}\nabla_\theta \bar{R}_\theta &= \nabla_\theta \sum_{\tau_{v_i}} P(\tau_{v_i}|\theta) R(\tau_{v_i}) \\ &= \sum_{\tau_{v_i}} P(\tau_{v_i}|\theta) \frac{\nabla_\theta P(\tau_{v_i}|\theta)}{P(\tau_{v_i}|\theta)} R(\tau_{v_i}) \quad (18) \\ &= \sum_{\tau_{v_i}} P(\tau_{v_i}|\theta) \nabla_\theta \log P(\tau_{v_i}|\theta) R(\tau_{v_i})\end{aligned}$$

From Eq. (18) we derived the expectation of $\bar{R}_\theta$, but the actual expectation $\sum_{\tau_{v_i}} P_\theta(\tau_{v_i}) \nabla_\theta \log P_\theta(\tau_{v_i}) R_i(\tau_{v_i})$ could not be derived, so we use sampling m trajectories and estimate the gradient by averaging the approximation:

$$\begin{aligned}\nabla_\theta \bar{R}_\theta &= \sum_{\tau_{v_i}} P(\tau_{v_i}|\theta) \nabla_\theta \log P(\tau_{v_i}|\theta) R(\tau_{v_i}) \\ &\approx \frac{1}{m} \sum_{i=1}^m \nabla_\theta \log P(\tau_{v_i}|\theta) R(\tau_{v_i}) \quad (19)\end{aligned}$$

then, we solve the $\nabla_\theta \log P(\tau_{v_i}|\theta)$ in:

$$\begin{aligned}\nabla_\theta \log P(\tau_{v_i}|\theta) &= \nabla_\theta \log \left[P(S_{v_i}(0)) \prod_{k=0}^H \frac{P(S_{v_i}(k+1)|S_{v_i}(k), A_{v_i}(k))}{\pi_\theta(A_{v_i}(k)|S_{v_i}(k))} \right] \\ &= \nabla_\theta \left[\frac{\log P(S_{v_i}(0)) + \sum_{k=0}^H \log P(S_{v_i}(k+1)|S_{v_i}(k), A_{v_i}(k))}{\sum_{k=0}^H \log \pi_\theta(A_{v_i}(k)|S_{v_i}(k))} \right] \quad (20) \\ &= \nabla_\theta \left[\sum_{k=0}^H \log \pi_\theta(A_{v_i}(k)|S_{v_i}(k)) \right] \\ &= \sum_{k=0}^H \nabla_\theta \log \pi_\theta(A_{v_i}(k)|S_{v_i}(k))\end{aligned}$$

In Eq. (20), only $\pi_\theta(A_{v_i}(k)|S_{v_i}(k))$ is relevant to agents, while the other terms are determined by the dynamics system, so the other terms can be removed. The final strategy gradient we get can be expressed as shown in Eq. (16).

B Hyperparameter Settings

The hyper-parameter settings in the experiment are shown in Table 4. Both the pressure level and the percentage of pressured agents in the GP are set to 0.5. The number of long-range neighbors and the preference in the neighbor selection in CNR are set to 1 and 0, respectively.

C Ablation Study of Model

We evaluated the influence of different model architectures on the performance of ACP under the same experimental setup. As shown in Table 5, the bidirectional LSTM (Bi-LSTM) model consistently outperformed both the MLP and unidirectional LSTM across all key metrics.

The superior performance of the Bi-LSTM aligns with its ability to capture complex non-linear and bidirectional dependencies in sequential data, enabling more effective aggregation of neighborhood

information and thus better fulfillment of both local and global objectives.

Table 4: Hyper-parameter settings of simulations.

Param.	Explanation	Value
T	Maximum time step	35
M	Maximum training round	300
ξ_l	Stabilization threshold (local goal)	0.25
ξ_g	Stabilization threshold (global goal)	0.5
r_c	Bounded confidence level	1
β	Merging threshold	0.5
μ	Opinion change threshold (experts)	0.5
ω	Consensus threshold	$1e^{-2}$
α	Learning rate	$1e^{-4}$
ϕ_{v_i}	Stubbornness level (non-experts)	[0.4, 0.6]
ϕ_{e_i}	Stubbornness level (experts)	[0.8, 0.9]
$s_{\dim}$	State input dimension in LSTM	4
$a_{\dim}$	Action output dimension in LSTM	1
$h_{\dim}$	Hidden dimension in LSTM	36
n_{layers}	Number of hidden layers in LSTM	2

Table 5: Ablation study of model architectures

Model	CC ↓	CS ↓	SCD ↓
MLP	1.72	33.63	0.11
LSTM	2.72	33.18	0.07
Bi-LSTM	1	23.72	0.05

D Analysis of Stubbornness in the Evolution of Opinions

We conducted experiments in an environment consisting of 40 non-experts and 4 experts. The opinion range was set between [0, 8], with the global goal U designated as 3. Given that the opinions of non-experts do not influence experts, we first analyzed the impact of stubbornness on the opinion evolution of experts. For ease of visualization, the stubbornness of non-experts was fixed at [0.4, 0.6], as shown in Figure 7. Subsequently, we fixed the stubbornness of experts at [0.8, 0.9] to examine the effect of stubbornness on the opinion evolution of non-experts, illustrated in Figure 8.

In Figure 7a, it is evident that when ϕ_{e_i} is too low, experts converge too rapidly to consensus. For experts on the periphery, their opinions change at an excessively fast rate, leading their followers to fall into local consensus without timely adjustment, thus converging slowly towards the target in subsequent time steps. Conversely, as shown in Figure 7d, when ϕ_{e_i} is too high, the slow pace of opinion changes among experts delays consensus, impeding the opinion evolution process for non-experts. Non-experts first align with experts and then gradually converge towards the target, resulting in a prolonged consensus period. Therefore, when ϕ_{e_i} is either too high or too low, experts exert a negative influence on the consensus of non-experts.

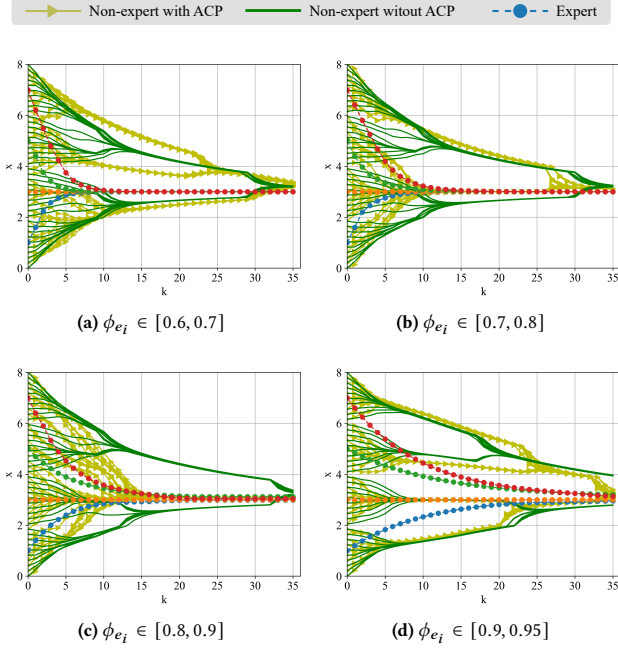


Figure 7: The impact of different stubbornness levels on experts' opinion evolution.

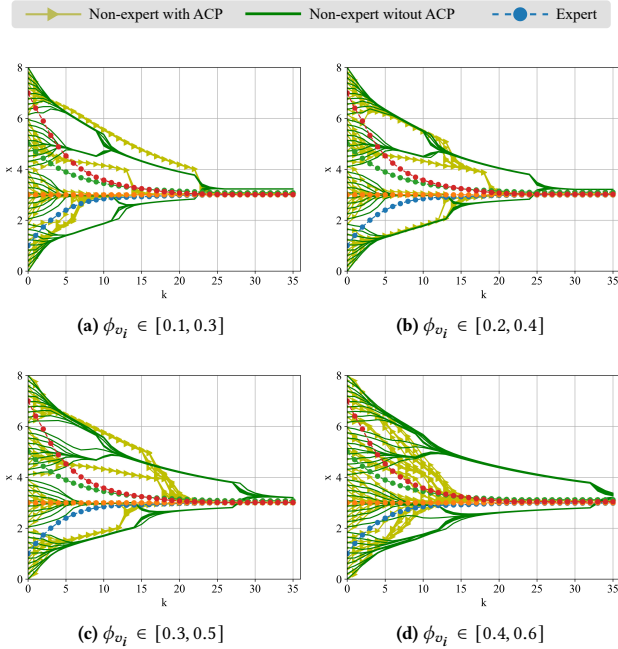


Figure 8: The impact of different stubbornness levels on non-experts' opinion evolution.

In contrast, when ϕ_{e_i} falls within a more appropriate range, as depicted in Figure 7b and Figure 7c, the rate of opinion change

among experts is moderate, thus positively influencing the consensus of non-experts. In Figure 7b, ϕ_{e_i} is lower compared to Figure 7c, guiding consensus for non-experts near the goal but causing peripheral non-experts to experience an island effect due to rapid expert opinion changes, losing potential communication partners. Our analysis reveals that when $\phi_{e_i} \in [0.8, 0.9]$, experts optimally facilitate the consensus of non-expert opinions.

From Figure 8a, it is evident that when ϕ_{v_i} is too low, the consensus speed of non-expert opinions significantly increases. This rapid consensus can lead to a strong echo chamber effect, causing non-experts to quickly fragment into multiple subgroups. These subgroups then slowly converge towards the goal under expert guidance. When the opinion divergence between any two subgroups falls within the bounded confidence range, these subgroups rapidly reach consensus.

As ϕ_{v_i} increases, as shown in Figures. 8b and 8c, the consensus speed of non-experts slows down. This reduction in consensus speed paradoxically promotes the achievement of global consensus. Initially, the formation of tight-knit subgroups among non-experts occurs more slowly, which means that during interactions between subgroups, they do not reach consensus too quickly but rather gradually converge towards the goal. In this scenario, subgroups positioned centrally act effectively as bridges for those at the periphery, preventing the occurrence of isolated clusters or “island effects.”

Ultimately, we find that when ϕ_{v_i} is within the interval $[0.4, 0.6]$ (Figure. 8d), the consensus quality and speed of non-experts are optimal. At this level of stubbornness, some non-experts serve as effective bridges between groups, ensuring that non-experts do not rapidly fall into local consensus but instead achieve efficient global consensus with the aid of ACP.

E Adversarial Agent Analysis

Social networks frequently contain malicious users who persistently disseminate harmful information. These adversarial agents maintain rigid opinions unaffected by interactions, purposefully propagating fixed viewpoints.

Although H-NeiFi lacks inherent adversarial detection, it safeguards global consensus convergence by blocking opinion propagation from malicious users during opinion exchange. Other agents receive fewer inputs from adversarial sources while malicious agents continue receiving normal opinions. This asymmetric filtering prevents malicious outputs from affecting benign users, thereby preserving consensus convergence despite persistent adversarial activity.

As shown in Figure 9, adversarial agent v_D maintains a fixed opinion ($x_{v_D}(k) = 3.5$). Results demonstrate v_D 's inability to disrupt global consensus convergence. H-NeiFi's filtering mechanism effectively isolates malicious influence while ensuring unimpeded consensus development among benign users.

F Analysis of Experts' Dynamic Influence

In this study, expert roles are defined as authoritative individuals, such as official government agency accounts, outstanding contributors in specific fields, or opinion leaders with large follower bases.

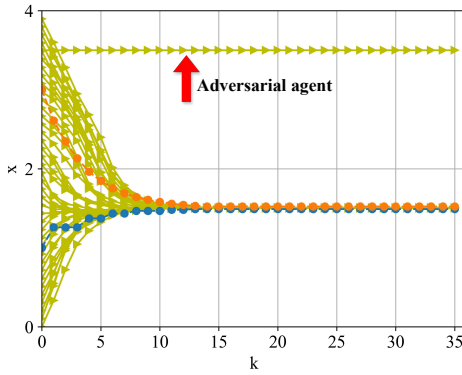


Figure 9: Evolution under adversarial influence.

The authoritativeness of these users is rooted in their social attributes and public recognition, rendering their status relatively stable and difficult to replace within a given time cycle.

To address the fluidity of influence, we propose periodically executing the H-NeiFi algorithm to adapt to potential shifts in user roles. This regular reassessment not only captures the evolution of social dynamics but also ensures the algorithm model remains synchronized with real-time public opinion environments. Furthermore, it enables flexible responses to changes in key figures across different periods, such as the emergence of new experts or fluctuations in the influence of existing ones.

G Opinion Evolution Analysis under Multi-Expert Conditions

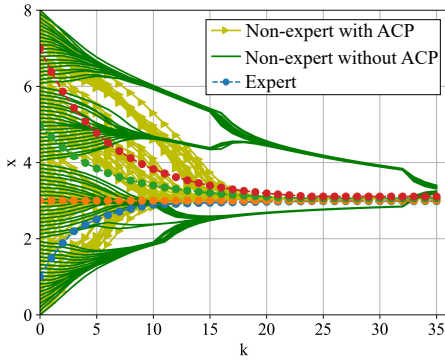


Figure 10: Opinion guidance processes in multi-expert settings.

Figure 10 illustrates the agents' evolutionary trajectory under the influence of multiple experts within the $[0, 8]$ opinion range. Initially, the experts' opinion distribution is $\{1, 3, 5, 7\}$, with the global consensus goal set to 3. It can be found that in the absence of

ACP interference, agents are constrained by r_c . Non-experts who initially follow the same expert first achieve internal consistency under the expert's influence. Even if some non-experts are initially closer to the global goal than the expert, they may move away from the goal due to the combined influence of the expert and their neighbors. After achieving local consensus, agents can only slowly converge to the global goal under expert guidance. This demonstrates that expert influence does not always accelerate consensus to the goal.

Under ACP's influence, both global and local goals are balanced using the local information perceived by the agents. When expert guidance hinders an agent's consensus toward the global goal, ACP compensates by adjusting neighborhood assignments to keep the agent on track. Additionally, considering the effect of bounded confidence, agents that converge too quickly can isolate others. For example, in the lower part of the green trajectory in the $[2, 4]$ range, two groups of agents directly reach the goal, but this causes the agents in the $[0, 2]$ range to lose their communicative "bridge" neighbors, leading to isolated clusters. This issue is avoided with ACP intervention, illustrating the importance of long-term planning. Peripheral agents must move quickly toward the goal, while those closer to the goal should act as "bridge" to maintain connectivity and prevent isolation during consensus.

H Ethical Considerations Discussion

The proposed opinion-guidance method demonstrates significant potential in facilitating rapid consensus formation by strategically shaping social interactions and communication patterns. This capability can enhance collaborative decision-making, particularly in contexts such as public health campaigns or crisis management, where unified public understanding is critical. However, the method could be exploited to unethically manipulate public opinion, with platforms potentially leveraging this technology for deceptive or coercive purposes, such as artificially amplifying specific experts to promote biased narratives or suppressing dissenting voices through selective communication barriers. Such misuse might manifest as deceptive political propaganda, commercial coercion, or the systematic marginalization of minority viewpoints, ultimately eroding trust in democratic processes and exacerbating social polarization. These risks highlight the urgent need for robust ethical safeguards to ensure the responsible deployment of the technology.

To prevent the negative impacts of malicious use, we advocate embedding human oversight across multiple dimensions: establishing independent ethics committees to review the selection of experts and prevent undue bias; creating transparent user agreements to safeguard users' autonomy and right to informed consent over their information ecosystems, allowing them to disable filtering functions and access unfiltered communication channels; and implementing regulated governance frameworks to ensure that global consensus goals align with public interest rather than corporate or political motives. Through these multidimensional safeguards, our method can be transformed from a tool of control into a catalyst for informed, inclusive public discourse.